

a standard Nokia handset with a 3.2-megapixel camera, developing a 'snap-on' addition that includes the microscope optics and a holder for blood samples on glass slides.

The CellScope uses cheap commercial light-emitting diodes as the light source—in place of the high-power, gas-filled lamps used in laboratory versions of the device—and cheap optical filters to isolate the light coming from the fluorescent tags. It has a resolution of just over one millionth of a metre.

Sensor to measure happiness

Scientists in the United States have created a sensor that is capable of measuring happiness of people.

Mathematician Peter Dodds and computer scientist Chris Danforth, working in the Advanced Computing Center at the University of Vermont, have developed a mechanism to measure happiness of millions of bloggers.

"The proliferation of personal online writing such as blogs gives us the opportunity to measure emotional levels in real time," they wrote in their study available in the online edition of *Journal of Happiness Studies*.

Their hedonometer gathers sentences beginning "I feel..." Then, by applying standardised psychological valence of words, each sentence receives a happiness score.

World's fastest supercomputer

A supercomputer named Novo-G, described by its lead designer as the most powerful computer of its kind in the world, has become operational at the University of Florida (UF).

Novo-G gets the first part of its name from the Latin term for "make anew, change, alter," and the second from 'G' for 'genesis.' A reconfigurable computer, it can rearrange its internal circuitry to suit the task at hand. Applications range from space satellites to research supercomputers—anywhere size, energy and high speed are important.

According to the university's website, traditional computers use

so-called fixed-logic devices to perform a large variety of tasks. This jack-of-all-trades approach requires a substantial amount of overhead in space and energy, no matter what work needs to be done. On the other hand, special-purpose computers can be built to perform certain tasks very well but are not flexible.

Reconfigurable computers make the best of both worlds. That is because they can rearrange their internal circuitry like Lego blocks, creating the most appropriate architecture for each assignment. As a result, a reconfigurable computer can be ten to 100 times faster than other computers of its size while using five to ten times less energy.

Although the concept has been proven, reconfigurable computers remain at the research stage and are not easy to use. One of the main goals of the NSF Center is to pioneer techniques to make reconfigurable computers more accessible.

Bacterial computers

Researchers in the US have developed bacterial computers that can revolutionise genetic research. The research team, including scientists from Missouri Western State University and Davidson College in North

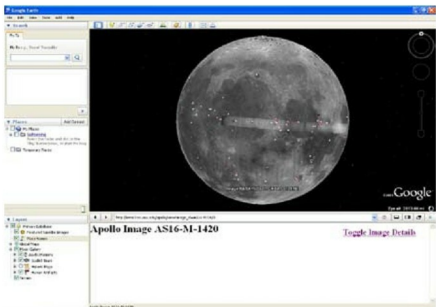
Carolina, USA, engineered the DNA of *Escherichia coli* bacteria and created bacterial computers capable of solving a classic mathematical problem known as the Hamiltonian Path problem.

The Hamiltonian Path problem asks whether there is a route in a network from a beginning node to an ending node, visiting each node exactly once. The researchers modified the genetic circuitry of the bacteria to enable them to find a Hamiltonian path in a three-node graph. Bacteria that successfully solved the problem reported their success by fluorescing both red and green, resulting in yellow colonies.

Discover the moon in Google Earth

Forty years ago, the crew of Apollo 11 walked on the moon. With Moon in Google Earth, it's now possible for you to follow in their footsteps.

Moon in Google Earth is an interactive 3D atlas of the Moon, viewable with Google Earth 5.0. It lets users explore a virtual Moonscape, follow guided tours from astronauts Buzz Aldrin (Apollo 11) and Jack Schmitt (Apollo 17), see the latest rover concepts by teams competing in the Google Lunar X-PRIZE, view high-resolution 'street view' style panoramic photos, watch previously unreleased



Moon in Google Earth